

Improving diagnostic yield in rare disease cohort through a novel structural variant analysis pipeline

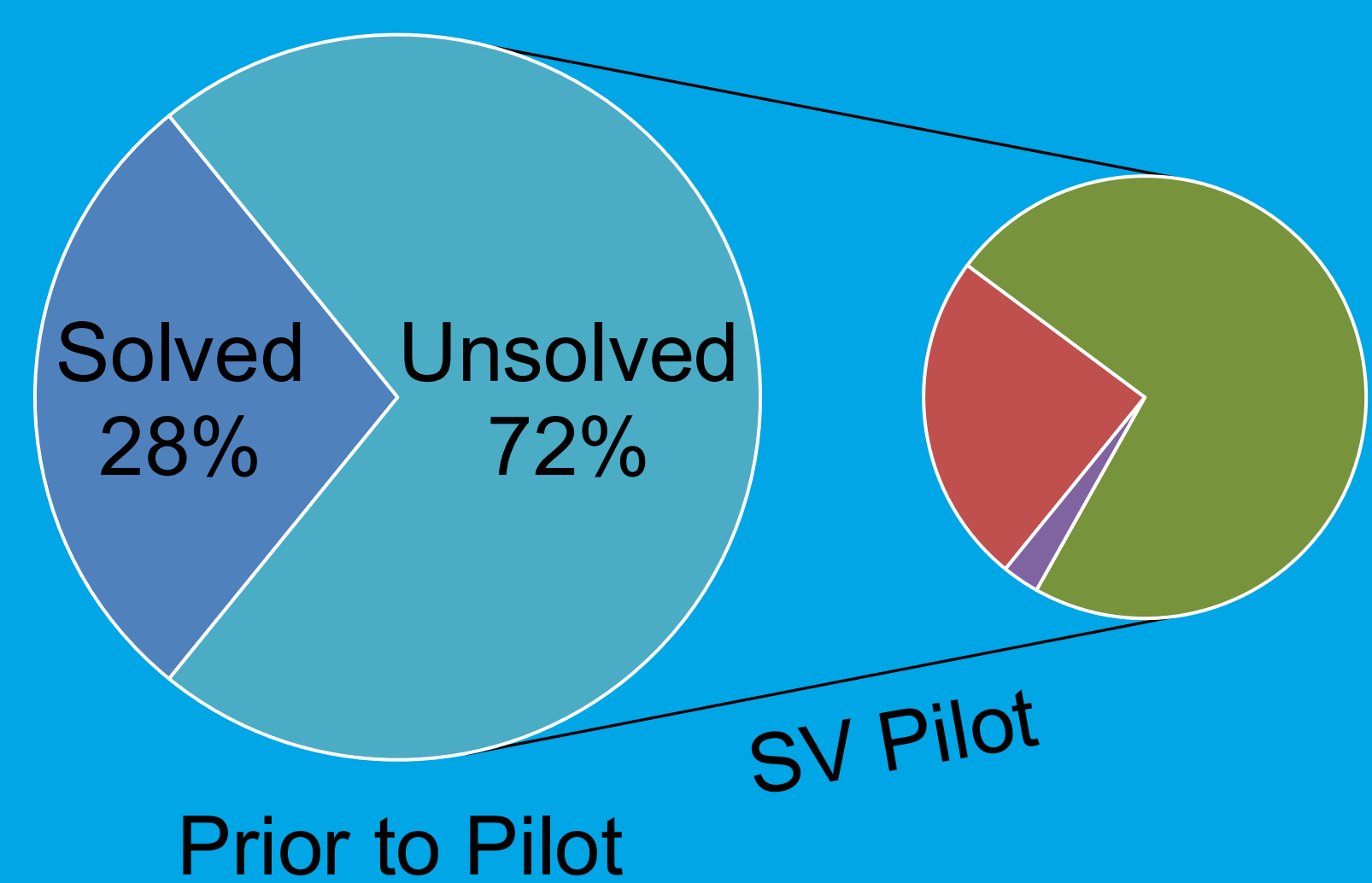
Shayne Plourde¹, Mohammad Marhabaie¹, Swetha Ramadesikan¹, Iftekhar A. Showpnil¹, Umamaheswaran Gurusamy¹, Allison Daley¹, Leeran Dublin-Ryan¹, Jesse M. Hunter¹, Peter White^{2,3}, Richard K. Wilson^{1,3}, Daniel C. Koboldt^{1,3}

¹The Steve and Cindy Rasmussen Inst. for Genomic Med., Nationwide Children's Hosp., Columbus, OH; ²Office of Data Sciences, Nationwide Children's Hosp., Columbus, OH; ³Dept. of Pediatrics, The Ohio State Univ. Coll. of Med., Columbus, OH

Highlights

- **Unsolved burden:** ~70% of families remain undiagnosed after SNV/indel/CNV analysis.
- **Scalable pipeline:** Developed an SV filtering and prioritization strategy which included a database of recurrent SV calls.
- **Efficiency gain:** Reduced review burden from >10,000 SVs per case to <30 candidates.
- **Recurrent Database:** Built a recurrent database of commonly observed Manta and GRIDSS SV calls.
- **Clinical impact:** Identified compelling SVs in 5 (10%) of the first 50 unsolved cases analyzed.

Rare Disease Cohort Diagnostic Yield with SV Analysis Pilot



- Solved
- Unsolved - SVs analyzed
- Unsolved SVs not analyzed
- SV Finding

Figure 1. Rare disease cohort yield for all 258 cases. From the unsolved cases 50 were selected to analyze SVs for and from those a candidate SV was found in 5.

Acknowledgments:

Unsolved Cases in Rare Disease Genome Analysis

- **Cohort:** 258 families with undiagnosed rare genetic disorders
- **Prior testing:** Many had negative exome or genome sequencing tests prior to enrolling.
- **Standard WGS analysis:** Identified SNVs, indels, and CNVs, solving ~28% of cases
- **Unresolved cases:** ~72% (185/258) remain without a diagnosis
- **Hypothesis:** Structural variants (SVs) may explain some of these unresolved cases

Structural Variant (SV) Analysis Pipeline

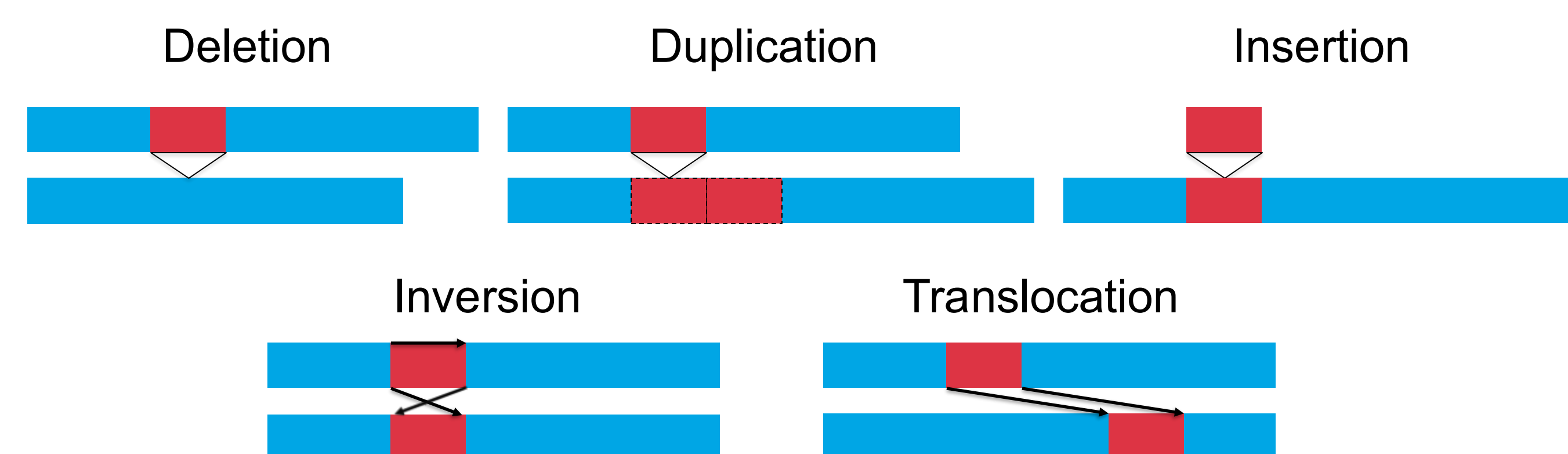


Figure 2. Structural variants detected using **Manta** and **GRIDSS**. Deletions, insertions, and duplications between 50 bp and 1 Mb, as well as translocations and inversions.

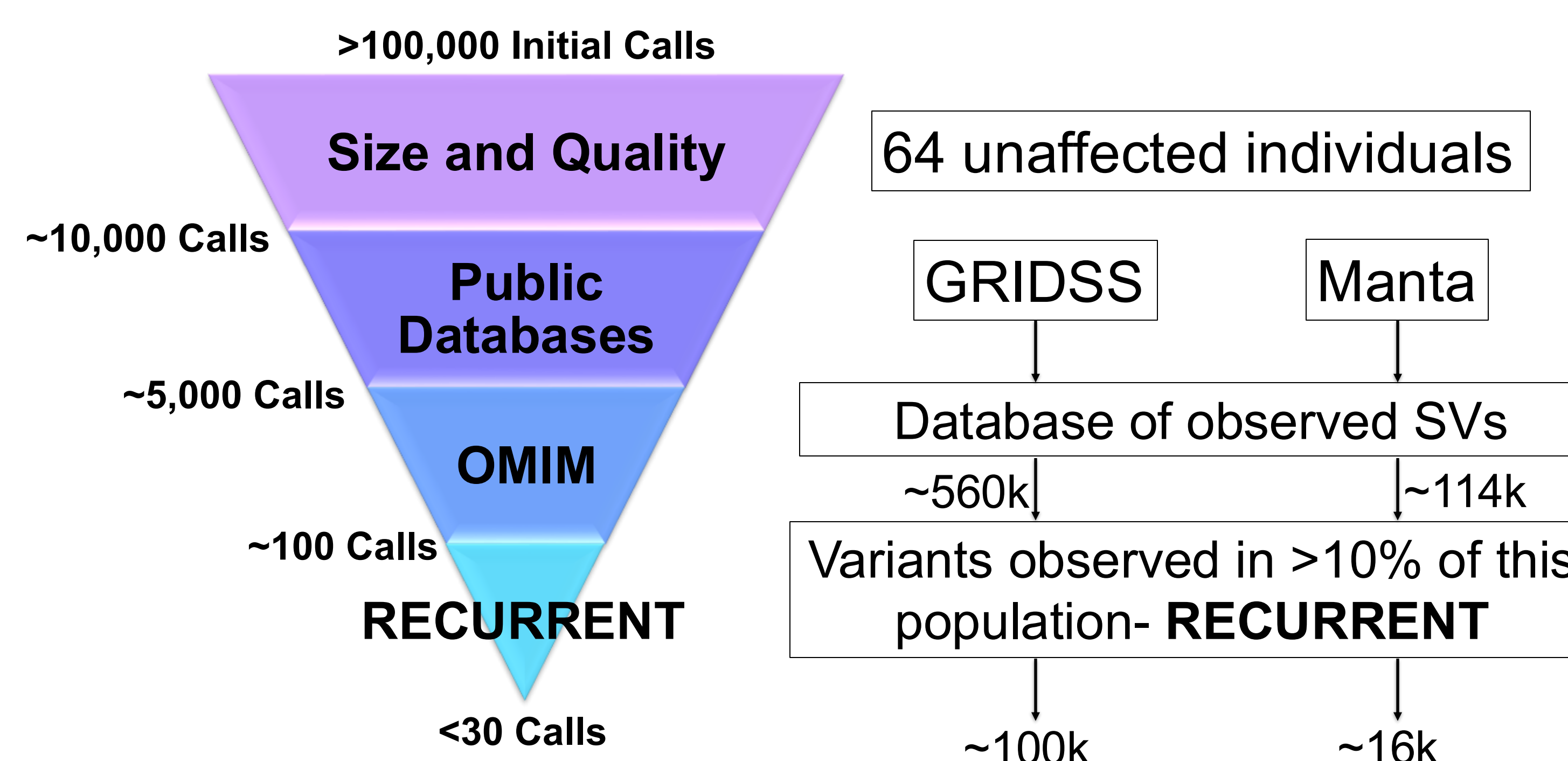


Figure 3. Filtering and prioritization for Manta and GRIDSS SV calls. This pipeline reduced the analyst review burden, enabling detection of 5 (10%) compelling candidate SVs in the first 50 unsolved cases.

FOXC1 de novo 329 bp deletion

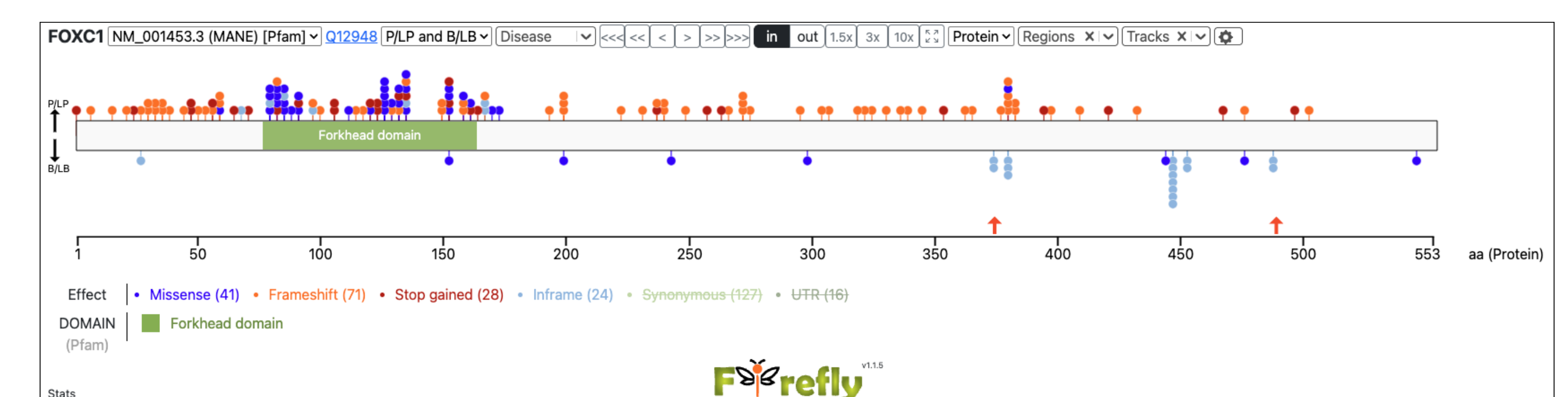
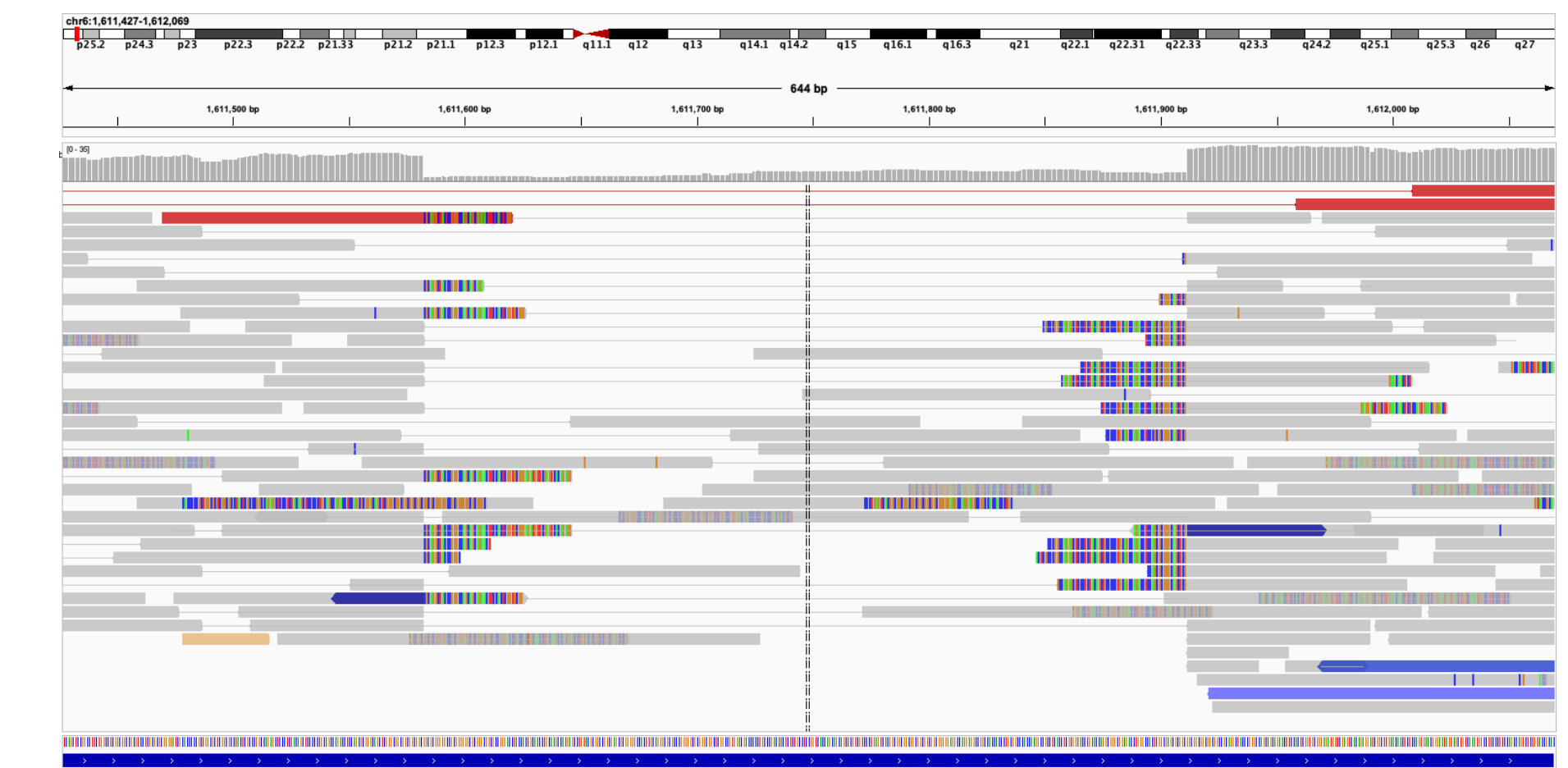


Figure 4. 3-year-old male with global developmental delay, bilateral clubfeet, congenital cardiac anomalies, dysmorphic facies, left sensorineural hearing loss, ptosis, optic nerve anomaly, and decreased vision. SV analysis identified a **de novo 329 bp deletion in FOXC1**. **FOXC1** variant cause **AXENFELD-RIEGER SYNDROME, TYPE 3; RIEG3**

SGCE de novo 182 kb inversion

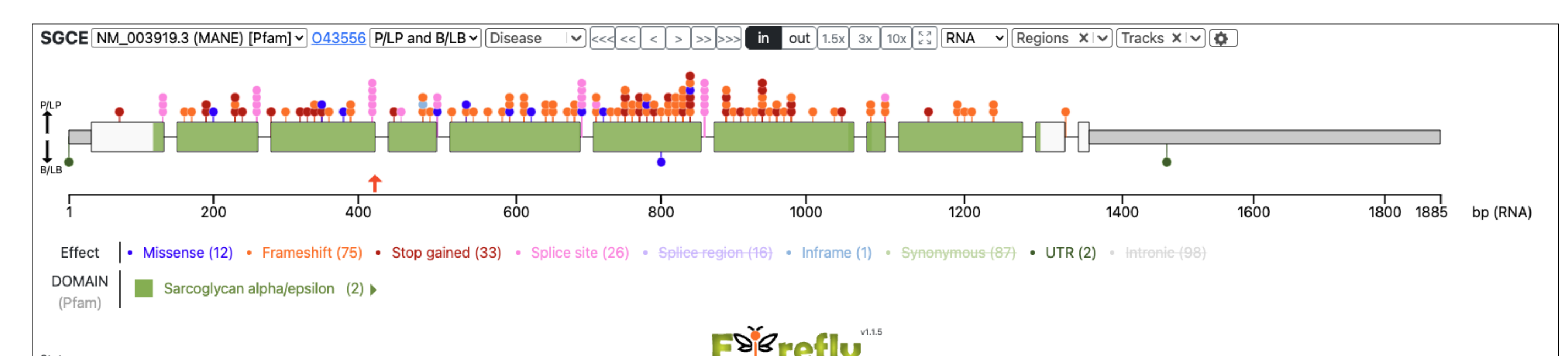
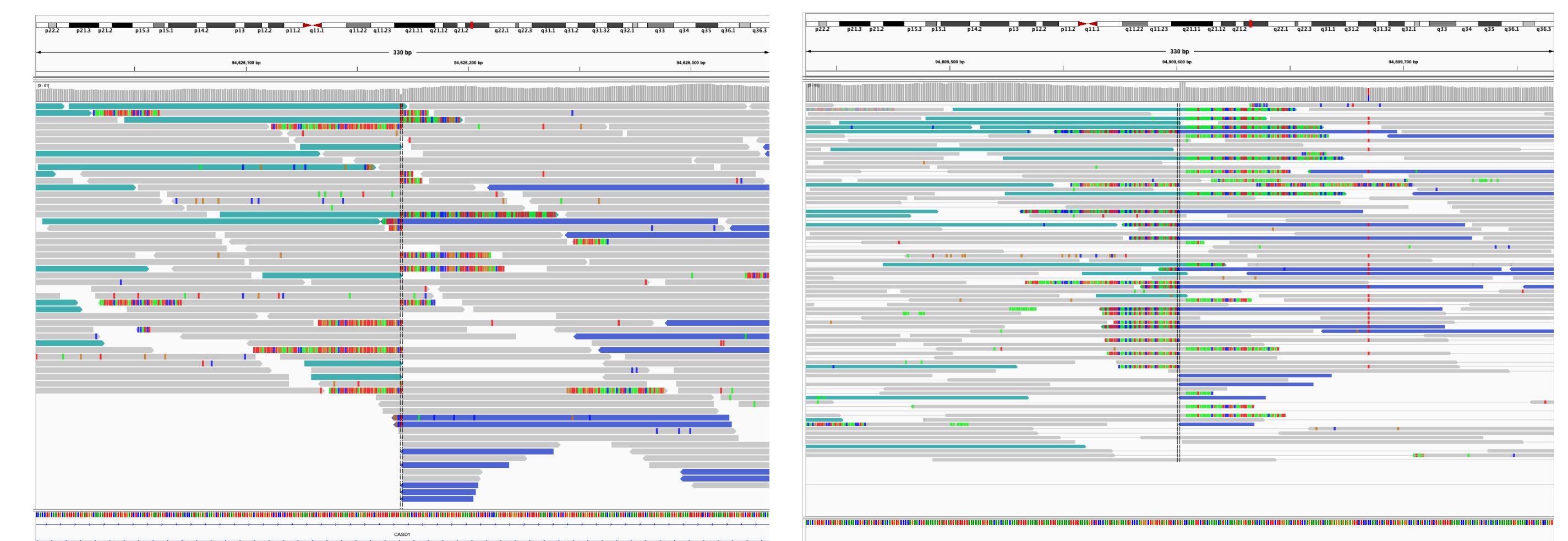
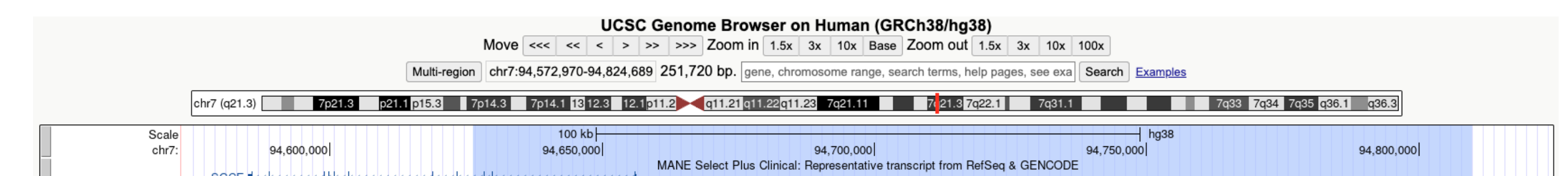


Figure 5. Female with a clinical diagnosis of primary myoclonus dystonia but no prior molecular finding. SV analysis identified a **de novo 182 kb inversion disrupting SGCE**. **SGCE** variants cause **DYSTONIA 11, MYOCLONIC; DYT11**.